

SPIKED SPECTRAL ACCELERATION FOR DEEP REINFORCEMENT LEARNING

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Paper under double-blind review

ABSTRACT

Value approximation plays a central role in deep reinforcement learning: small errors in the value estimator can be amplified through repeated bootstrapping and policy-value updates, degrading stability and sample efficiency. Recent work using random matrix theory has shown that the singular-value spectrum of network weights evolves from a Marchenko–Pastur (MP)–like bulk at initialization toward a bulk-plus-spikes structure as training uncovers task signal. Motivated by this spectral view, we propose Spike Spectral Acceleration Regularization (SSAR), a lightweight regularizer that encourages candidate spike singular values to separate from the MP bulk earlier in training. By driving the spectrum toward the Baik–Ben Arous–Péché (BBP) transition sooner, SSAR helps value networks capture task-relevant structure earlier and produce more reliable training targets for the policy. We provide theoretical analysis showing that SSAR reduces value-estimation error under reasonable assumptions, and we derive a practical stopping criterion based on deviation from the initial MP distribution. Empirically, SSAR improves sample efficiency, stability, and final performance on challenging continuous-control benchmarks compared to common baselines.

1 INTRODUCTION

Value networks are fundamental to deep reinforcement learning because they supply the scalar estimates of future return that guide policy improvement and carry reward information back to earlier decisions. In practice these value functions are almost always implemented with deep neural networks, which are powerful but imperfect: with limited interactions and limited computing budget they produce biased and noisy estimates. The training regime in reinforcement learning makes this especially challenging. Targets for the value network are computed from the agent’s current behavior and its own estimators, and value updates are interleaved with policy updates rather than performed in isolation. In many settings each data point can only be used for a small number of gradient steps before new data arrives or the policy changes, so the value network rarely has the chance to fully converge between updates. As a result, small errors in the value estimate tend to reappear in subsequent targets and can grow over time through the coupled update process Tang & Berseth (2024). The practical consequences are straightforward: inaccurate value estimates can steer the policy toward worse actions, slow the backward propagation of reward information to early states, and reduce sample efficiency.

Prior work attacks this problem from several directions. $TD(\lambda)$ reduces reliance on one-step bootstrapped targets by forming multi-step returns, which helps propagate reward information more reliably Sutton et al. (1998). Double Q-learning and other ensemble methods address overestimation and variance in the target values Van Hasselt et al. (2016); Hasselt (2010); Lee et al. (2021). Using a delayed copy of the network as a target network also stabilizes learning by smoothing the bootstrapping targets Mnih et al. (2013). These techniques are widely adopted and effective in practice, but don’t actually focus on the neural value approximator itself, which involve the dynamic of deep neural networks.

Recent work shows that random matrix theory provides a clear and intuitive way to understand how neural networks organize information during training. At the start, the singular values of a network’s weight matrices typically follow the Marchenko–Pastur (MP) law, which is exactly what we expect from a purely random initialization. As training proceeds, however, the spectrum gradually drifts

away from this random baseline. A small set of singular values begin to grow noticeably larger than the rest, indicating that the network is starting to identify directions that are aligned with the task. In other words, the weight matrix begins to split into two distinct parts: a wide, noisy bulk of small singular values and a much smaller set of strong, high-energy directions that carry task-relevant information. This evolution generally unfolds in stages: an early period that still appears mostly random, a middle phase where signal gradually pulls away from noise, and a later stage in which the spectrum often develops a heavy-tailed shape as more structured, correlated patterns are learned. The middle phase, where signal separates from noise, is almost unavoidable and represents a crucial turning point. It is the point that the network starts to encode meaningful information about the problem it is solving. The speed of this separation plays a crucial role in learning, since a faster split allows the network to use meaningful signal earlier, leading to quicker learning and more reliable convergence, while a slower split delays progress and makes learning less efficient.

Due to the limited training budget for value networks in reinforcement learning, signal and noise can remain entangled for a considerable portion of training, which in turn allows sizable estimation errors to propagate into subsequent policy updates. Motivated by this insight and inspired by results from random matrix theory such as the Baik–Ben Arous–Péché (BBP) phase transition Baik et al. (2005), we propose a regularization method that explicitly accelerates the separation between signal-related and noise-related singular values. By encouraging this separation to occur earlier in training, the value network can capture task-relevant structure sooner and provide more reliable targets to the policy, leading to faster, more stable learning.

Concretely, we introduce Spike Spectral Acceleration Regularization (SSAR). SSAR monitors the singular value spectrum during the middle stages of training and adds a lightweight loss term that encourages candidate spike singular values to move away from the Marchenko–Pastur (MP) bulk. In practice, this causes the BBP phase transition to occur earlier than it would under standard training, so the network can pick up task-relevant signal sooner. Our main contributions are as follows. (1) We propose SSAR, a simple and principled regularizer guided by random matrix theory that speeds up signal–noise separation in value networks. (2) We provide theoretical analysis showing that SSAR lowers value-estimation error under reasonable assumptions, and we derive a practical stopping rule based on deviations from the MP distribution. (3) We empirically validate SSAR on challenging continuous-control benchmarks and show consistent gains in sample efficiency, stability, and final performance compared to common baselines. (4) We release our code to support reproducibility and fair comparisons.

2 RELATED WORK

Spectral analysis of matrices that arise in neural networks has recently attracted a lot of interest as a way to understand what these models learn and to suggest practical improvements for training Pappan (2018); Sagun et al. (2016); Ge et al. (2021); Pennington & Worah (2017). The Marchenko–Pastur (MP) law, first described in Marčenko & Pastur (1967), captures the singular-value distribution one expects from a random initialization of a layer. Building on that baseline, Martin and Mahoney Martin & Mahoney (2021) and follow-up works have traced how the empirical spectrum changes during training and identified a recurring “5+1” phase transition: the spectrum starts MP-like, a few singular values peel away to form a bulk-plus-spikes structure, and with further training the spectrum often develops heavy-tailed behavior as correlated structure grows. Other papers have used these spectral signatures in different ways — for example, using heavy-tailedness to predict generalization Martin et al. (2021), probing the role of small singular values in large language models Staats et al. (2024), or pruning by treating singulars below the MP edge as noise Berlyand et al. (2023; 2025). Most prior work focuses on late-stage spectra or post hoc analysis of trained models; the early-stage spectral changes, when signal first peels away from noise, have received far less attention even though they directly affect how quickly a network can learn. To our knowledge, this paper is the first to study that early-stage evolution specifically for value-network training and to propose a practical method for speeding up the signal–noise separation during training.

3 PRELIMINARIES AND NOTATION

We consider a Markov decision process (MDP) given by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \mu, \gamma)$, where $\mathcal{S}$ and $\mathcal{A}$ are the state and action spaces, $P(s' | s, a)$ is the transition kernel, $R(s, a) \in \mathbb{R}$ is the reward function, μ is the initial state distribution, and $\gamma \in (0, 1)$ is the discount factor. The goal is to learn a stochastic policy $\pi^\phi(a | s)$, parameterized by ϕ , that maximizes the expected discounted return

$$J(\pi^\phi) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right], \quad (1)$$

where actions are sampled $a_t \sim \pi^\phi(\cdot | s_t)$ and transitions follow $s_{t+1} \sim P(\cdot | s_t, a_t)$.

For a fixed policy π^ϕ , the state–action value function (Q-function) $Q^\theta : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, parameterized by θ , is defined as

$$Q^\theta(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid (s_0, a_0) = (s, a) \right]. \quad (2)$$

The optimal Q-function Q^* is the fixed point of the Bellman optimality operator $\mathcal{T}$:

$$\mathcal{T}Q(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a'} Q(s', a') \right]. \quad (3)$$

Fitted Q-iteration and related fitted-Q algorithms attempt to approximate this fixed point by minimizing a mean-squared Bellman residual over a distribution $\mathcal{D}$ of transitions (s, a, s') . Define the Bellman residual

$$r(s, a) = Q^\theta(s, a) - \mathcal{T}Q^\theta(s, a),$$

and the corresponding loss

$$L_0(Q^\theta) = \mathbb{E}_{(s, a, s') \sim \mathcal{D}} \left[\frac{1}{2} r(s, a)^2 \right] = \mathbb{E}_{(s, a, s') \sim \mathcal{D}} \left[\frac{1}{2} (Q^\theta(s, a) - \mathcal{T}Q^\theta(s, a))^2 \right]. \quad (4)$$

A standard gradient-based update (in fitted-Q style) is

$$\theta' = \theta - \eta \nabla_\theta L_0(Q^\theta) = \theta - \eta \mathbb{E}_{(s, a, s') \sim \mathcal{D}} [r(s, a) \nabla_\theta Q^\theta(s, a)], \quad (5)$$

where $\eta > 0$ denotes the step size.

To connect our spectral analysis to network parameters, consider a typical parametrization of the Q-network where $W \in \mathbb{R}^{n \times m}$ denotes the penultimate-layer weight matrix, $o \in \mathbb{R}^m$ the final-layer weight vector, $h(s, a) \in \mathbb{R}^n$ the input to the penultimate layer, $\psi(\cdot)$ the penultimate-layer activation, and $z(s, a) = \psi(Wh(s, a)) \in \mathbb{R}^m$ the penultimate-layer output. The Q-function can then be written as $Q^\theta(s, a) = o^\top z(s, a)$, and we take θ to collect all network parameters (including W and o).

For the matrix $W \in \mathbb{R}^{n \times m}$ we use its singular value decomposition (SVD). Let $\{s_k, u_k, v_k\}_{k \geq 1}$ denote the singular triplets with singular values ordered as $s_1 \geq s_2 \geq \dots \geq 0$. The unit-norm left and right singular vectors $u_k \in \mathbb{R}^n$ and $v_k \in \mathbb{R}^m$ satisfy

$$Wv_k = s_k u_k, \quad W^\top u_k = s_k v_k. \quad (6)$$

We recall two standard results from random matrix theory that we use in the paper.

Theorem 3.1 (Marchenko–Pastur law Marčenko & Pastur (1967)). *Let X be an $n \times m$ random matrix with i.i.d. entries having zero mean and variance $\sigma^2 < \infty$. Define the sample covariance $Y = \frac{1}{m} X X^\top$ and let $\rho_Y(\lambda)$ be the empirical spectral distribution (ESD) of Y . If $n, m \rightarrow \infty$ with $n/m \rightarrow c \in (0, \infty)$, then almost surely the ESD ρ_Y converges weakly to the Marchenko–Pastur distribution*

$$\rho_Y^c(\lambda) = \frac{1}{2\pi c \sigma^2} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda} \mathbf{1}_{[\lambda_-, \lambda_+]},$$

where $\lambda_\pm = \sigma^2(1 \pm \sqrt{c})^2$ are the edges of the support.

Theorem 3.2 (BBP phase transition Baik et al. (2005)). *Let $X \in \mathbb{C}^{n \times m}$ have i.i.d. complex Gaussian columns with population covariance matrix $\Sigma = \text{diag}(\ell_1, \dots, \ell_n)$, and form $Y = \frac{1}{m} X X^\top$. Fix an integer $r > 0$ and suppose that $\ell_{r+1} = \dots = \ell_N = 1$, while $\ell_1, \dots, \ell_r$ are the top population eigenvalues of covariance matrix. Assume $n, m \rightarrow \infty$ with $n/m \rightarrow c \in [1, \infty)$. If for some*

0 $\leq k \leq r$, $\ell_1 = \dots = \ell_k = 1 + c^{\frac{1}{2}}$, and the remaining spikes $\ell_{k+1}, \dots, \ell_r$ lie in $(0, 1 + c^{\frac{1}{2}})$, then the largest sample eigenvalue λ_1 satisfies $\mathbb{P}\left(\frac{(\lambda_1 - (1 + c^{\frac{1}{2}})^2)}{c^{-\frac{1}{2}}(1 + c^{-\frac{1}{2}})^{4/3}} m^{2/3} \leq x\right) \rightarrow F_k(x)$, as $n, m \rightarrow \infty$. If for some $1 \leq k \leq r$ the top population spikes satisfy $\ell_1 = \dots = \ell_k \in (1 + c^{\frac{1}{2}}, \infty)$, then λ_1 satisfies $\mathbb{P}\left(\frac{(\lambda_1 - (1 + \frac{\ell_1 c}{\ell_1 - 1})) \cdot \sqrt{m}}{\sqrt{\ell_1^2 - \frac{\ell_1^2 c}{(\ell_1 - 1)^2}}} \leq x\right) \rightarrow G_k(x)$, where F_k is a Tracy–Widom-type Fredholm-determinant distribution, and G_k is a finite-dimensional Gaussian-type limit, precisely defined in Baik et al. (2005). This theorem is also extended from the largest eigenvalue of Y to several eigenvalues Benaych-Georges & Nadakuditi (2011).

The BBP phenomenon describes how spiked population eigenvalues ℓ_i affect the behavior of the sample eigenvalues of Y . Roughly speaking, if a population spike ℓ_i does not exceed the critical threshold $1 + \sqrt{c}$, then the corresponding sample eigenvalue remains at the bulk edge and exhibits Tracy–Widom-type fluctuations; if $\ell_i > 1 + \sqrt{c}$, then the corresponding sample eigenvalue separates from the bulk (a “spike”) and its fluctuations are asymptotically Gaussian.

4 ASSUMPTIONS

Assumption 1 (Smoothness). *The baseline loss $L_0(Q_\theta)$ is L -smooth in parameters (in particular in W): for all W, W' , $\|\nabla_W L_0(W) - \nabla_W L_0(W')\|_F \leq L \|W - W'\|_F$*

Consequently the descent lemma holds:

$$L_0(W + \Delta) \leq L_0(W) + \langle \nabla_W L_0(W), \Delta \rangle + \frac{L}{2} \|\Delta\|_F^2.$$

Assumption 2 (Step-size regime). *Fix a constant $0 < \theta < 1$. The step-size $\eta > 0$ is small enough so that $L\eta \leq \frac{1}{2}$. We will choose the spike weight $\gamma > 0$ later in the Theory section; here we only require the step-size smallness to control second-order terms.*

5 METHOD

We propose a lightweight spectral-acceleration module that can be added to standard value-network training to encourage earlier separation of signal singular values from the noise bulk. In this section we first state a formal proposition that summarizes the relevant Random matrix theory behavior for a spike model, then explain the motivation, define the spike regularizer, give a practical stop rule, and present the complete algorithm.

Proposition 5.1. *Consider the spike model $X = R + P \in \mathbb{R}^{n \times m}$, where R is a random matrix with i.i.d. entries of zero mean and variance σ^2 , and $P = \sum_{i=1}^r \theta_i u_i v_i^\top$ is a fixed-rank perturbation with singular strengths θ_i and population directions u_i, v_i . Let the aspect ratio satisfy $n, m \rightarrow \infty$ with $n/m \rightarrow c \in (0, \infty)$, and denote by $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ the eigenvalues of $\frac{1}{m} X X^\top$. There exists a critical spike level θ_c such that:*

- If $\theta_i \leq \theta_c$ for all $i \in [1, r]$, then $\lambda_1 = \lambda_+ + O(m^{-2/3})$, where $\lambda_+ = \sigma^2(1 + \sqrt{c})^2$ is the right edge of the MP bulk.
- If some $k \leq r$ spikes satisfy $\theta_i > \theta_c$ for $i \in [1, k]$, then those k sample eigenvalues separate from the bulk and satisfy $\lambda_i = f(\theta_i) + O(m^{-1/2})$ for $i \in [1, k]$, where $f(\theta_i) > \lambda_+$.

The proposition is proved in the Appendix and follows from Theorem 3.2 and results in Benaych-Georges & Nadakuditi (2011).

Motivation We view the target-layer weight matrix during training as $W = W_{\text{init}} + \Delta W$, where W_{init} behaves like a random matrix R and ΔW encodes the learned, low-rank task signal $\sum_{i=1}^r \theta_i u_i v_i^\top$. At initialization the empirical spectrum of $\frac{1}{m} W_{\text{init}} W_{\text{init}}^\top$ follows the Marchenko–Pastur law (Theorem 3.1). According to Proposition 5.1, when signal strengths are below the BBP threshold the largest sample eigenvalue remains near the MP edge λ_+ , up to $O(m^{-2/3})$ fluctuations. Once one or more signal directions exceed the threshold, the corresponding sample

eigenvalues “jump” away from the bulk to values $f(\theta_i) > \lambda_+$, up to $O(m^{-1/2})$ fluctuations. This jump marks the formation of a spectral spike, i.e., the separation of signal from noise. Our key idea is to accelerate this spike formation so that the network captures task-relevant directions earlier in training, which in turn yields more reliable value estimates under the tight optimization budgets typical in RL.

Spike regularizer Guided by the above, we design a simple per-update regularizer that targets near-edge singular modes. Let λ_+ denote the right edge of the MP bulk for the background ensemble of W . Fix a constant $c > 0$ and define the edge-window width $T = c m^{-2/3}$, and the spectral edge window $I = [\lambda_+, \lambda_+ + T]$. We call an empirical singular value s_k a *near-edge* mode if $s_k^2/m \in I$, which means squared singular value normalized by m (eigenvalue of $\frac{1}{m}WW^T$) lies inside the edge window. For each such mode we add a per-mode penalty $(T - s_k^2/m)_+$, where $(x)_+ = \max\{x, 0\}$. The multi-mode spike regularizer is $L_{\text{spike}}(W) = \sum_{k: s_k^2/m \in I} (T - s_k^2/m)_+$.

Given the standard Bellman residual loss $L_0(Q_\theta)$, the total training loss becomes $L_{\text{tot}}(\theta) = L_0(Q_\theta) + \gamma L_{\text{spike}}(W)$, where $\gamma > 0$ is a small regularization weight. In practice the spike module computes the leading singular values of the target-layer weight matrix W via SVD.

Stopping rule Our spike model is most appropriate during the early-to-middle training stages. As stated in related work, Martin & Mahoney (2021) analyze the network training and develop a theory to identify 5+1 phases of training, in later stages the spectrum may develop strong correlations and heavy-tailed structure where the spike interpretation no longer applicable. To avoid applying SSAR past its regime of validity, we use a simple *soft-rank* metric $R_s(W) = \frac{\lambda_+}{\lambda_{\max}}$, where $\lambda_{\max} = s_{\max}^2/m$ is the largest sample eigenvalue. Empirically, a small R_s indicates that one or more spikes have already separated from the bulk; we therefore apply SSAR only while $R_s(W) > R_c$ for some threshold $R_c \in (0, 1)$.

Algorithm The SSAR module is applied on top of standard actor-critic or fitted-Q updates. Algorithm 1 summarizes the high-level training loop and Algorithm 2 gives the SSAR loss computation that runs every e_{spike} episodes (or training iterations).

In practice, SSAR is inserted into the normal training loop as follows. Every e_{spike} iterations we extract the target-layer weight matrix W , compute its leading singular value $s_{\max}$ (and a set of top singulars s_i), and set $\lambda_{\max} = s_{\max}^2/m$ and $\lambda_+ = \sigma^2(1 + \sqrt{n/m})^2$. We then form the soft-rank $R_s = \lambda_+/\lambda_{\max}$ and compare it to a chosen threshold R_c . If $R_s > R_c$ (indicating the spectrum is still in the early/middle stage), we define the edge window $I = [\lambda_+, \lambda_+ + c, m^{-2/3}]$ with $T = c, m^{-2/3}$, compute the SSAR loss L_{SSAR} and add γL_{SSAR} to the usual Bellman loss $L_0(Q_\theta)$. The value-network parameters are then updated by gradient descent on the combined loss, and the policy is updated as usual. If $R_s \leq R_c$, the SSAR term is skipped for that iteration. This procedure repeats until convergence. The next section presents theoretical guarantees and empirical results.

6 PERFORMANCE ANALYSIS

We study the effect of adding a single-spike spectral regularizer to the loss for a two-layer network. Let $W \in \mathbb{R}^{m \times n}$ denote the first-layer weight matrix, hidden width m , input dimension n . We assume the matrix admits the low-rank plus noise decomposition $W = Z + \sum_{i=1}^r \theta_i u_i v_i^\top$, where $u_i \in \mathbb{R}^m$, $v_i \in \mathbb{R}^n$ are unit vectors, $\theta_i \in \mathbb{R}$ are fixed signal strengths and r is fixed. The network output at input $h \in \mathbb{R}^n$ is $Q^\theta(h) = o^\top \varphi(Wh)$, where $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ acts coordinatewise and is differentiable on the domain of interest.

Focus on a single (top) empirical singular triple (s, u, v) of W with $Wv = su$ and $\|u\|_2 = \|v\|_2 = 1$. Define the single-spike regularizer $L_{\text{spike}}(W) = (T_{\text{thr}} - s(W)^2)_+$; when the spike is active (i.e. $s^2 < T_{\text{thr}}$) the subgradient is $g_{\text{spike}}(W) = \nabla_W L_{\text{spike}}(W) = -2s uv^\top$.

We compare the vanilla and spike-augmented single-step updates with step-size $\eta > 0$. The vanilla parameter difference is $\Delta_{\text{van}} = -\eta \nabla_W L_0(W)$, while the spike-augmented difference is $\Delta_{\text{sp}} = -\eta (\nabla_W L_0(W) + \gamma g_{\text{spike}}(W))$. Define the one-step Bellman-loss improvement gap

Algorithm 1 SSAR (high-level)

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1: Input: policy params  $\phi$ , value params
    $\theta$ , hyperparams,  $e_{\text{spike}}, \gamma$ 
2: Initialize  $\phi, \theta$  (zero-mean, variance
    $\sigma^2$ ), set episode  $e \leftarrow 0$ 
3: repeat
4:    $e \leftarrow e + 1$ 
5:   Compute  $L_{SSAR} \leftarrow \text{COMPUTE-SSAR-LOSS}(\theta, e)$ 
6:    $\theta \leftarrow \theta - \eta \nabla_{\theta}(L_0(Q^{\theta}) + \gamma L_{SSAR})$ 
7:    $\phi \leftarrow \phi - \eta \nabla_{\phi} L_0(\pi^{\phi})$ 
8: until convergence

```

Algorithm 2 Compute-SSAR-Loss

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1: Input:  $\theta, \sigma, c, R_c, e, e_{\text{spike}}$ 
2:  $L_{SSAR} \leftarrow 0$ 
3: if  $e \bmod e_{\text{spike}} \neq 0$  then
4:   return  $L_{SSAR}$ 
5: end if
6: Extract target-layer weight  $W$  from  $\theta$ 
7: Compute top singular values  $s_1, \dots, s_r$  (and  $s_{\max}$ )
8:  $\lambda_+ \leftarrow \sigma^2(1 + \sqrt{n/m})^2$ ,  $\lambda_{\max} \leftarrow s_{\max}^2/m$ 
9: if  $\lambda_+/\lambda_{\max} < R_c$  then
10:  return 0  $\triangleright$  stop SSAR (spikes already
    separated)
11: end if
12:  $T \leftarrow c m^{-2/3}$ ,  $I \leftarrow [\lambda_+, \lambda_+ + T]$ 
13: for each  $s_i$  in  $s_1, \dots, s_r$  do
14:   if  $s_i^2/m \in I$  then
15:      $L_{SSAR} \leftarrow L_{SSAR} + (T - s_i^2/m)_+$ 
16:   end if
17: end for
18: return  $L_{SSAR}$ 

```

$\Delta_t := (L_0(W + \Delta_{\text{sp}}) - L_0(W)) - (L_0(W + \Delta_{\text{van}}) - L_0(W))$; negative Δ_t means the spike update decreases L_0 more than the vanilla update.

To quantify alignment between the spike direction and the Bellman gradient, set $d(h) := o \circ \varphi'(Wh) \in \mathbb{R}^m$ (elementwise product). By the chain rule $\nabla_W Q^{\theta}(h) = d(h) h^{\top}$ and $\nabla_W L_0(W) = \mathbb{E}[r(h) d(h) h^{\top}]$. Define the alignment scalar $\alpha := \langle -\nabla_W L_0(W), uv^{\top} \rangle = \mathbb{E}[r(h) (u^{\top} d(h)) (h^{\top} v)]$. Let $v^* \in \mathbb{R}^n$ denote the population signal direction and decompose $v = \tau_n v^* + v_{\perp}$ with $\tau_n := \langle v, v^* \rangle$ and $\langle v_{\perp}, v^* \rangle = 0$. Set $T := h^{\top} v^*$, $U(T) := \mathbb{E}[u^{\top} d(h) | T]$, and $M := \mathbb{E}[r(h) U(T) T]$, $A := \tau_n M$.

We now give the a lemma and theorem to show the performance improvement of our method. The proof and the assumption used in the proof will show in Appendix A.

Lemma 6.1 (Alignment lower bound). *For any fixed $\vartheta \in (0, 1)$ there exists $M_0(\vartheta)$ such that for all $m \geq M_0(\vartheta)$ we have $\alpha \geq (1 - \vartheta) A$.*

Theorem 6.1 (Single-spike single-step improvement). *When the spike is active as the $g_{\text{spike}} = -2suw^{\top}$ applies, choose step-size $\eta = \kappa A$ with $\kappa > 0$ sufficiently small, then there exists $c_0 = \frac{3}{4}\kappa s(1 - \vartheta) > 0$ such that for all sufficiently large m , $\Delta_t \leq -c_0 \gamma A^2$.*

As shown in 6.1, the spike-augmented update yields an immediate extra population-level decrease in the Bellman loss of order $\Theta(\gamma A^2)$ per step. Intuitively A is the population-level alignment transmitted through the population direction v^* . The key mechanism is the linear cross-term $-\eta \lambda \langle \nabla_W L_0, \nabla_W L_{\text{spike}} \rangle$ in the one-step expansion: Lemma 6.1 ensures this cross-term has a definite negative sign and size comparable to the population signal A , while higher-order terms can be controlled by choosing η sufficiently small.

7 EXPERIMENTS

In this section, we validate our proposed SSAR algorithm using some high-dimensional continuous control tasks from the DeepMind control suite (Tassa et al., 2018). We compare SSAR with baseline algorithm (without spike accelerated) and with L1 or L2 regularization. What’s more, we make some ablation experiment: the influence of the size of acceleration window, the comparison of acceleration of penultimate and multiple layers, the influence of different ”stop point” in stop rule.

- Synthetic spiked-model simulation to verify the predicted $\Omega(\gamma n^{-1/3})$ scaling in the near-critical regime.
- Controlled RL environments (e.g. small gridworld / control tasks) with instrumented penultimate-layer SVD to show accelerated formation of task-relevant directions and improved sample efficiency.
- Ablations: varying γ , step-size schedules, and number of accelerated modes.

Detailed empirical protocols, datasets, and plots will be added in the experimental section.

8 DISCUSSION AND LIMITATIONS

We emphasize the assumptions and limitations: the main theorem requires existence of a near-critical signal (BBP regime), and the present analysis is at population level (finite-sample/algorithmic noise and practical approximations are left for future work). We also discuss possible extensions: multi-signal generalizations, adaptive choices of γ , and combining with regularization to avoid pathological behavior in practice.

9 CONCLUSION

Summary and directions for future work.

AUTHOR CONTRIBUTIONS

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

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A BBP TRANSITION

A.1 FROM BBP DISTRIBUTIONAL LIMITS TO O_p -SCALE CONCLUSIONS

NOTATION

Let $M, N \rightarrow \infty$ with $M/N \rightarrow \gamma^2 \in (0, \infty)$. Denote the largest sample eigenvalue by λ_1 , the Marchenko–Pastur right edge by λ_+ , and for a supercritical population spike $\ell_1 > \ell_c$ denote by $\theta(\ell_1)$ the deterministic outlier location solving the self-consistent equation. We write $X_n = O_p(a_n)$ to mean X_n/a_n is tight.

Lemma A.1 (Distributional convergence $\Rightarrow O_p$). *Let X_n be random variables, and let $a_n > 0$, $b_n \in \mathbb{R}$ be deterministic sequences. If*

$$a_n(X_n - b_n) \xrightarrow{d} Y,$$

where Y is a non-degenerate random variable (i.e. not a point mass), then

$$X_n - b_n = O_p(1/a_n).$$

Proof. By the definition of convergence in distribution, for any fixed $K > 0$ we have

$$\lim_{n \rightarrow \infty} \Pr(|a_n(X_n - b_n)| > K) = \Pr(|Y| > K).$$

Fix $\varepsilon > 0$. Choose K large enough that $\Pr(|Y| > K) < \varepsilon/2$. Then there exists N such that for all $n \geq N$,

$$\Pr(|a_n(X_n - b_n)| > K) \leq \Pr(|Y| > K) + \varepsilon/2 < \varepsilon.$$

Set $C := K$. Then for $n \geq N$,

$$\Pr(|X_n - b_n| > C/a_n) = \Pr(|a_n(X_n - b_n)| > C) < \varepsilon,$$

which is exactly the statement $X_n - b_n = O_p(1/a_n)$. $\square$

APPLICATION TO BBP CASE (A): EDGE / TW-TYPE SCALING

BBP states (case (a)) that, for the appropriate constants,

$$A_M(\lambda_1 - b_M) \xrightarrow{d} F_k,$$

where in the notation used in Baik–Ben Arous–Péché one has

$$A_M = \frac{\gamma}{(1 + \gamma)^{4/3}} M^{2/3}, \quad b_M = (1 + \gamma^{-1})^2 = \lambda_+,$$

and F_k is a non-degenerate limit law (Tracy–Widom-type parameterized family). Applying Lemma A.1 with $X_n = \lambda_1$, $a_n = A_M$, $b_n = \lambda_+$ yields

$$\lambda_1 - \lambda_+ = O_p(1/A_M) = O_p(M^{-2/3}).$$

Thus

$$\lambda_1 = \lambda_+ + O_p(M^{-2/3})$$

as required.

APPLICATION TO BBP CASE (B): OUTLIER / CLT-TYPE SCALING

BBP states (case (b)) that, after centering by a deterministic sequence c_M and scaling by $B_M \sim \sqrt{M}$,

$$B_M(\lambda_1 - c_M) \xrightarrow{d} G_k,$$

where G_k is a non-degenerate limit law (finite-dimensional CLT-type), and c_M is the explicit BBP center (in the large- M limit c_M equals $\theta(\ell_1)$ up to $o(1)$ terms). Applying Lemma A.1 with $a_n = B_M$, $X_n = \lambda_1$, $b_n = c_M$ gives

$$\lambda_1 - c_M = O_p(1/B_M) = O_p(M^{-1/2}).$$

It remains to connect c_M with the deterministic outlier location $\theta(\ell_1)$. Standard finite-rank perturbation theory (and the deterministic equivalents used in BBP/Benaych–Georges) yield

$$c_M = \theta(\ell_1) + o(1).$$

Consequently,

$$\lambda_1 - \theta(\ell_1) = (\lambda_1 - c_M) + (c_M - \theta(\ell_1)) = O_p(M^{-1/2}) + o(1) = O_p(M^{-1/2}).$$

Therefore

$$\lambda_1 = \theta(\ell_1) + O_p(M^{-1/2})$$

as required.

B APPENDIX: ASSUMPTIONS AND PROOFS OF MAIN THEOREM

In this section, We first give some new assumptions, a new lemma and its proof is presented. Finally, we will show the proof of lemma 6.1 and theorem 6.1.

Assumption

B1 (Input: sub-Gaussian, bounded covariance). The input $h \in \mathbb{R}^n$ satisfies $\mathbb{E}[h] = 0$ and for every fixed $a \in \mathbb{R}^n$ the projection $\langle a, h \rangle$ is sub-Gaussian with $\|\langle a, h \rangle\|_{\psi_2} \leq K_h \|a\|_2$. The covariance $\Sigma_h = \text{Cov}(h)$ satisfies $\lambda_{\max}(\Sigma_h) \leq C_\Sigma$. The reward is bounded.

B2 (Spike model for W). The first-layer weight matrix admits a fixed low-rank plus noise decomposition

$$W = Z + \sum_{i=1}^r \theta_i u_i v_i^\top,$$

where r does not grow with m, n , each $u_i \in \mathbb{R}^m$, $v_i \in \mathbb{R}^n$ are unit vectors, and $\theta_i \in \mathbb{R}$ are deterministic. The noise matrix $Z \in \mathbb{R}^{m \times n}$ has independent Gaussian rows with $\mathbb{E}[Z] = 0$ and $\text{Cov}(\text{vec}(Z))$ having bounded trace; concretely we assume each row $z_j^\top$ of Z is Gaussian with $\mathbb{E}[z_j] = 0$ and $\text{Cov}(z_j) = \Sigma_z$ and the rows z_j are independent across j . The signal part couples rows only through the common right singular vectors v_i (rank- r signal).

B3 (Activation regularity and output layer scaling). The activation φ is C^2 on the domain of interest, $\sup_x |\varphi'(x)| \leq B_0$, and φ' is L -Lipschitz. Consequently (taking $\varphi(0)$ finite) we have the linear-growth bound $|\varphi(x)| \leq |\varphi(0)| + B_0|x|$. The output vector o and the left singular vector u satisfy $\max_j |o_j| \leq C_o/m$ and $\max_j u_j^2 \leq C_u/m$ for constants C_o, C_u independent of m, n .

B4 (Structural Bellman dependence – signal through T). The Bellman/residual structure satisfies the low-dimensional conditional-mean assumption: there exists a measurable function $G : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\mathbb{E}[r(h) \mid h] = G(T), \quad T = h^\top v^*.$$

In words: the conditional expectation of the residual given h depends on h only through the scalar projection T .

B5 (Non-degeneracy of main signal). Define $U(T) := \mathbb{E}[u^\top d(h) \mid T]$ and $M := \mathbb{E}[G(T)U(T)T]$. There exists $m_0 > 0$ such that $M \geq m_0$.

Remark B.1. All constants $K_h, K_z, C_\Sigma, C_o, C_u, B_0, L, C_{\xi,2}, C_{\xi,4}, C_{g,4}, m_0$ are model constants that do not scale with m, n (except where indicated).

B1 is widely used in proof in some reinforcement learning paper. B2 is the assumption of spike model. B3 is the assumption of activation function and the normalization of of output layer. B4 encodes the structural modeling assumption that the effective signa in the Bellman residual is carried by the scalar projection $T = h^\top v^$.*

B.1 LEMMA (FOURTH-MOMENT CONTROL FOR $Q, \mathcal{T}Q, r$ AND G)

Lemma B.1 (Fourth-moment control). *Under Assumptions B1–B5, the following hold:*

1. *There exists a finite constant C_Q such that*

$$\mathbb{E}[Q^\theta(h)^4] \leq C_Q.$$

2. *There exists $C_{\mathcal{T}Q} < \infty$ such that*

$$\mathbb{E}[(\mathcal{T}Q^\theta(h))^4] \leq C_{\mathcal{T}Q}.$$

3. *Consequently the Bellman residual $r(h) = Q^\theta(h) - \mathcal{T}Q^\theta(h)$ satisfies $\mathbb{E}[r(h)^4] \leq C_r < \infty$ for some finite C_r , and the function $G(T) = \mathbb{E}[r \mid T]$ satisfies $\mathbb{E}[G(T)^4] \leq C_G < \infty$.*

Proof. We start the proof by serval steps:

Step 1: per-coordinate fourth moments of $\varphi(w_j^\top h)$. Write the j th row of W as $w_j^\top = z_j^\top + \sum_{i=1}^r \theta_i u_{i,j} v_i^\top$ (so $w_j^\top h = z_j^\top h + \sum_{i=1}^r \theta_i u_{i,j} T_i$ with $T_i := v_i^\top h$). By B1 and the Gaussianity of z_j (B2), the scalar $z_j^\top h$ is sub-Gaussian with variance parameter bounded by $\lambda_{\max}(\Sigma_z) \lambda_{\max}(\Sigma_h)$ (hence all moments exist). The signal term $\sum_{i=1}^r \theta_i u_{i,j} T_i$ is a linear combination of the random variables $T_i = v_i^\top h$, each of which is sub-Gaussian by B1; coefficients $\theta_i u_{i,j}$ are deterministic and bounded because $\|u_i\|_2 = 1$ and θ_i are fixed. Therefore $w_j^\top h$ has uniformly bounded moments of all orders: in particular there exists $C_{w,4}$ such that

$$\mathbb{E}[(w_j^\top h)^4] \leq C_{w,4}, \quad \text{uniformly in } j.$$

Using the linear-growth bound $|\varphi(x)| \leq |\varphi(0)| + B_0|x|$ (B3), we obtain

$$|\varphi(w_j^\top h)|^4 \leq C(1 + |w_j^\top h|^4)$$

for some C depending on $|\varphi(0)|, B_0$. Taking expectation and using the previous bound gives

$$\mathbb{E}[\varphi(w_j^\top h)^4] \leq C_\varphi < \infty,$$

uniformly in j .

Step 2: control of $\mathbb{E}[\|\varphi(Wh)\|_2^4]$. We have

$$\|\varphi(Wh)\|_2^4 = \left(\sum_{j=1}^m \varphi(w_j^\top h)^2 \right)^2 = \sum_{j=1}^m \varphi(w_j^\top h)^4 + 2 \sum_{1 \leq i < j \leq m} \varphi(w_i^\top h)^2 \varphi(w_j^\top h)^2.$$

Take expectation. For the diagonal terms,

$$\sum_{j=1}^m \mathbb{E}[\varphi(w_j^\top h)^4] \leq mC_\varphi.$$

For the off-diagonal terms we use Cauchy–Schwarz:

$$\mathbb{E}[\varphi(w_i^\top h)^2 \varphi(w_j^\top h)^2] \leq \sqrt{\mathbb{E}[\varphi(w_i^\top h)^4] \mathbb{E}[\varphi(w_j^\top h)^4]} \leq C_\varphi.$$

Hence

$$\mathbb{E}[\|\varphi(Wh)\|_2^4] \leq mC_\varphi + 2 \binom{m}{2} C_\varphi = C_\varphi (m + m(m-1)) \leq C_\varphi m^2.$$

Thus there exists $C_{\varphi,4}$ such that

$$\mathbb{E}[\|\varphi(Wh)\|_2^4] \leq C_{\varphi,4} m^2.$$

Step 3: fourth moment of $Q^\theta(h)$. Recall $Q^\theta(h) = \sum_{j=1}^m o_j \varphi(w_j^\top h)$ and the Cauchy–Schwarz bound

$$|Q^\theta(h)| \leq \|o\|_2 \|\varphi(Wh)\|_2.$$

By B3, $\max_j |o_j| \leq C_o/m$, hence

$$\|o\|_2^2 = \sum_{j=1}^m o_j^2 \leq m \cdot \frac{C_o^2}{m^2} = \frac{C_o^2}{m},$$

so

$$\|o\|_2^4 \leq \frac{C_o^4}{m^2}.$$

Combining with Step 2 yields

$$\mathbb{E}[Q^\theta(h)^4] \leq \|o\|_2^4 \mathbb{E}[\|\varphi(Wh)\|_2^4] \leq \frac{C_o^4}{m^2} \cdot (C_{\varphi,4} m^2) = C_o^4 C_{\varphi,4},$$

which is a finite constant independent of m .

Step 4: fourth moment of $\mathcal{T}Q^\theta(h)$. The Bellman operator acts by conditional expectation over next-state and reward, so for every h

$$\mathcal{T}Q^\theta(h) = \mathbb{E}[R(h, \xi) + \gamma Q^\theta(h') \mid h],$$

where R is the one-step reward and h' the next observation. Using Jensen’s inequality and the tower property,

$$\mathbb{E}[(\mathcal{T}Q^\theta(h))^4] \leq \mathbb{E}[\mathbb{E}[(R(h, \xi) + \gamma Q^\theta(h'))^4 \mid h]] = \mathbb{E}[(R + \gamma Q^\theta)^4],$$

where the right-hand side is the full fourth moment of the one-step return plus discounted next-value. Reward is bounded according to B1 and the bound on $\mathbb{E}[Q^\theta(h)^4]$ from Step 3, the right-hand side is finite. Hence there exists $C_{\mathcal{T}Q} < \infty$ with the desired bound.

Step 5: fourth moment of $r(h)$ and of $G(T)$. Since $r(h) = Q^\theta(h) - \mathcal{T}Q^\theta(h)$, by Minkowski (or simply by the polynomial expansion and Cauchy–Schwarz) we get

$$\mathbb{E}[r(h)^4] \leq C(\mathbb{E}[Q^\theta(h)^4] + \mathbb{E}[(\mathcal{T}Q^\theta(h))^4]) \leq C_r < \infty.$$

Finally, $G(T) = \mathbb{E}[r \mid T]$ is a conditional expectation; by Jensen’s inequality,

$$\mathbb{E}[G(T)^4] = \mathbb{E}[(\mathbb{E}[r \mid T])^4] \leq \mathbb{E}[\mathbb{E}[r^4 \mid T]] = \mathbb{E}[r^4] \leq C_r,$$

so $G(T)$ has finite fourth moment as well.

This completes the proof of Lemma B.1. $\square$

B.2 PRELIMINARY MOMENT FACTS

We collect two elementary consequences of the assumptions (used repeatedly below).

Proposition B.1 (Moments of T and r). *Under B1 and B4, there exist finite constants C_T, C_r (depending only on K_h and $C_{g,4}, C_{\xi,4}$) such that $\mathbb{E}[T^4] \leq C_T$ and $\mathbb{E}[r^4] \leq C_r$.*

Proof. By B1 the projection $T = \langle v^*, h \rangle$ is sub-Gaussian with $\|T\|_{\psi_2} \leq K_h \|v^*\|_2 = K_h$. Standard sub-Gaussian moment bounds imply $\mathbb{E}[T^4] \leq C_T$ for some C_T depending only on K_h . For $r = g(T) + \xi$ we have by the elementary inequality $|a + b|^4 \leq 8(|a|^4 + |b|^4)$ that $\mathbb{E}[r^4] \leq 8(\mathbb{E}[g(T)^4] + \mathbb{E}[\xi^4]) \leq 8(C_{g,4} + C_{\xi,4}) =: C_r$. $\square$

B.3 LEMMA (VARIANCE CONTROL).

Lemma B.2. *Define the conditional residual $\zeta(h) := u^\top d(h) - U(T)$ so that $\mathbb{E}[\zeta \mid T] = 0$. Under B1–B3, there exists a constant C_{var} depending only on the model constants such that $\mathbb{E}[\text{Var}(u^\top d(h) \mid T)] \leq C_{\text{var}}/m$. Equivalently, $\mathbb{E}[\zeta(h)^2] \leq C_{\text{var}}/m$.*

Proof. We proceed in steps.

Step 1 (row-wise decomposition and conditional independence). From the matrix spike model $W = Z + \sum_{i=1}^r \theta_i u_i v_i^\top$ we get the j -th row $w_j^\top = z_j^\top + \sum_{i=1}^r \theta_i u_{i,j} v_i^\top$, where $u_{i,j}$ is the j -th entry of u_i . For any fixed h , the scalar $w_j^\top h$ decomposes as $w_j^\top h = z_j^\top h + \sum_{i=1}^r \theta_i u_{i,j} (v_i^\top h)$. The deterministic-in- z_j term $\sum_i \theta_i u_{i,j} (v_i^\top h)$ is measurable w.r.t. $\sigma(h)$, while $z_j^\top h$ is a mean-zero sub-Gaussian scalar and the collection $\{z_j^\top h\}_{j=1}^m$ is independent across j by B2. Hence, conditional on h , the random variables $X_j := \varphi'(w_j^\top h) = \varphi'(a_j(h) + z_j^\top h)$ are independent across j , where $a_j(h) := \sum_{i=1}^r \theta_i u_{i,j} (v_i^\top h)$ is deterministic given h .

Step 2 (conditional variance expansion). For $S(h) := u^\top d(h) = \sum_{j=1}^m u_j o_j X_j$, conditional on h the independence of $\{X_j\}$ yields $\text{Var}(S(h) \mid h) = \sum_{j=1}^m u_j^2 o_j^2 \text{Var}(X_j \mid h)$ because cross-terms vanish.

Step 3 (bound single-term conditional variances). By B3 φ' is L -Lipschitz, hence for any two real numbers y_1, y_2 we have $|\varphi'(y_1) - \varphi'(y_2)| \leq L|y_1 - y_2|$. Thus $\text{Var}(X_j \mid h) = \text{Var}(\varphi'(a_j(h) + z_j^\top h) \mid h) \leq L^2 \text{Var}(z_j^\top h \mid h)$. Since $\text{Var}(z_j^\top h \mid h) = h^\top \Sigma_z h \leq \lambda_{\max}(\Sigma_z) \|h\|_2^2$, we deduce $\text{Var}(X_j \mid h) \leq L^2 \lambda_{\max}(\Sigma_z) \|h\|_2^2$.

Step 4 (expectation of conditional variance). Taking expectation in Step 2 and using Step 3,

$$\mathbb{E}[\text{Var}(S(h) \mid h)] \leq L^2 \lambda_{\max}(\Sigma_z) \mathbb{E}[\|h\|_2^2] \sum_{j=1}^m u_j^2 o_j^2.$$

By B1 $\mathbb{E}[\|h\|_2^2] = \text{tr}(\Sigma_h) \leq n \lambda_{\max}(\Sigma_h)$, and by B3 we have $\max_j u_j^2 \leq C_u/m$ and $\max_j o_j^2 \leq C_o^2/m$. Hence $\sum_{j=1}^m u_j^2 o_j^2 \leq m \cdot (C_u/m) \cdot (C_o^2/m) = C_u C_o^2/m$. Therefore $\mathbb{E}[\text{Var}(S(h) \mid h)] \leq C^{(1)}/m$ with $C^{(1)} := L^2 \lambda_{\max}(\Sigma_z) \text{tr}(\Sigma_h) C_u C_o^2$.

Step 5 (law of total variance relative to T). We apply the law of total variance with the refinement $\sigma(h) \supseteq \sigma(T)$: $\text{Var}(S(h) \mid T) = \mathbb{E}[\text{Var}(S(h) \mid h) \mid T] + \text{Var}(\mathbb{E}[S(h) \mid h] \mid T)$. Taking full expectation yields $\mathbb{E}[\text{Var}(S(h) \mid T)] = \mathbb{E}[\text{Var}(S(h) \mid h)] + \mathbb{E}[\text{Var}(\mathbb{E}[S(h) \mid h] \mid T)]$. The first term is bounded by $C^{(1)}/m$ from Step 4.

Step 6 (bound the second term). Define $\psi(h) := \mathbb{E}[S(h) \mid h] = \sum_{j=1}^m u_j o_j \mu_j(h)$ where $\mu_j(h) := \mathbb{E}_{z_j}[\varphi'(a_j(h) + z_j^\top h)]$. We bound $\mathbb{E}[\text{Var}(\psi(h) \mid T)] \leq \mathbb{E}[\psi(h)^2]$. Using Cauchy–Schwarz, $\psi(h)^2 \leq (\sum_{j=1}^m u_j^2 o_j^2) (\sum_{j=1}^m \mu_j(h)^2)$. By B3 $|\varphi'| \leq B_0$, so $|\mu_j(h)| \leq B_0$ and $\sum_{j=1}^m \mu_j(h)^2 \leq m B_0^2$. Together with $\sum_j u_j^2 o_j^2 \leq C_u C_o^2/m$ we obtain $\psi(h)^2 \leq C_u C_o^2 B_0^2/m$ and hence $\mathbb{E}[\psi(h)^2] \leq C^{(2)}/m$ for $C^{(2)} := C_u C_o^2 B_0^2$.

Step 7 (conclusion). Combining steps 4 and 6, $\mathbb{E}[\text{Var}(S(h) \mid T)] \leq (C^{(1)} + C^{(2)})/m := C_{\text{var}}/m$. Recalling $S(h) = u^\top d(h)$ and $\zeta(h) = S(h) - U(T)$ gives the stated result. $\square$

B.4 LEMMA (ALIGNMENT DECOMPOSITION AND QUANTITATIVE BOUND).

Then we give the precise definition of lemma 6.1:

Lemma B.3. Define $\alpha = \langle -\nabla_W L_0, uv^\top \rangle = \mathbb{E}[r(h)(u^\top d(h))(h^\top v)]$, $U(T) = \mathbb{E}[u^\top d(h) \mid T]$ and $M := \mathbb{E}[g(T)U(T)T]$, $A := \tau_n M$. Then the exact algebraic identity $\alpha = A - \varepsilon_n$ holds with $\varepsilon_n = -\tau_n \mathbb{E}[r\zeta T] - \mathbb{E}[r(u^\top d)(h^\top v_\perp)]$. Moreover there exists a constant C_{tot} (depending only on the model constants and C_{var}) such that $\frac{|\varepsilon_n|}{A} \leq \frac{C_{\text{tot}}}{m_0} \cdot \frac{1}{\tau_n \sqrt{m}}$. Finally get the result showed in lemma 6.1 $\alpha \geq (1 - \vartheta) A$

Proof. The proof has two parts: (i) an exact algebraic decomposition; (ii) quantitative bounds for the error terms.

Part (i) Algebraic decomposition. Using $v = \tau_n v^* + v_\perp$ and $u^\top d = U(T) + \zeta$ we expand

$$\alpha = \mathbb{E}[r(U + \zeta)(\tau_n T + h^\top v_\perp)].$$

Expanding the product yields four terms. The term $\mathbb{E}[rU(h^\top v_\perp)]$ vanishes because, conditioning on T , $h^\top v_\perp$ has mean zero (since $v_\perp \perp v^*$ and h has mean zero and covariances bounded by B1), and $U(T)$ is $\sigma(T)$ -measurable while $r = g(T) + \xi$ with $\mathbb{E}[\xi \mid T] = 0$. Thus only three terms remain and regrouping gives $\alpha = \tau_n \mathbb{E}[rUT] + \tau_n \mathbb{E}[r\zeta T] + \mathbb{E}[r\zeta(h^\top v_\perp)]$. Rearranging yields the stated identity with $\varepsilon_n = -\tau_n \mathbb{E}[r\zeta T] - \mathbb{E}[r(u^\top d)(h^\top v_\perp)]$.

Part (ii) Quantitative bound. Split $\varepsilon_n = \varepsilon^{(1)} + \varepsilon^{(2)}$ according to the two terms above.

Bound for $\varepsilon^{(1)} = -\tau_n \mathbb{E}[r\zeta T]$. By Cauchy-Schwarz $|\mathbb{E}[r\zeta T]| \leq \sqrt{\mathbb{E}[r^2 T^2]} \sqrt{\mathbb{E}[\zeta^2]}$. From Proposition A.1 we have $\mathbb{E}[r^4] \leq C_r$ and $\mathbb{E}[T^4] \leq C_T$, hence $\mathbb{E}[r^2 T^2] \leq \sqrt{C_r C_T} =: \tilde{C}_{rT}$. By Lemma B.2 $\mathbb{E}[\zeta^2] \leq C_{\text{var}}/m$. Therefore $|\varepsilon^{(1)}| \leq \tau_n \sqrt{\tilde{C}_{rT} C_{\text{var}}/m}$. Dividing by $A = \tau_n M \geq \tau_n m_0$ (B5) yields $\frac{|\varepsilon^{(1)}|}{A} \leq \frac{\sqrt{\tilde{C}_{rT} C_{\text{var}}}}{m_0 \sqrt{m}}$.

Bound for $\varepsilon^{(2)} = -\mathbb{E}[r(u^\top d)(h^\top v_\perp)]$. By Cauchy-Schwarz $|\varepsilon^{(2)}| \leq \sqrt{\mathbb{E}[r^2 (u^\top d)^2]} \sqrt{\mathbb{E}[(h^\top v_\perp)^2]}$. We bound the two factors separately.

(1) For $\sqrt{\mathbb{E}[(h^\top v_\perp)^2]}$ note $\mathbb{E}[(h^\top v_\perp)^2] = v_\perp^\top \Sigma_h v_\perp \leq \lambda_{\max}(\Sigma_h) \|v_\perp\|_2^2 \leq \lambda_{\max}(\Sigma_h)$. Put $C_h := \sqrt{\lambda_{\max}(\Sigma_h)}$.

(2) For $\sqrt{\mathbb{E}[r^2 (u^\top d)^2]}$ use Hölder: $\mathbb{E}[r^2 (u^\top d)^2] \leq \sqrt{\mathbb{E}[r^4]} \sqrt{\mathbb{E}[(u^\top d)^4]} \leq \sqrt{C_r} \sqrt{\mathbb{E}[(u^\top d)^4]}$. It remains to show $\mathbb{E}[(u^\top d)^4] \leq C_{ud}/m^2$ for some C_{ud} depending only on model constants.

Write $u^\top d = \sum_{j=1}^m c_j Y_j$ with $c_j := u_j o_j$ and $Y_j := \varphi'(w_j^\top h)$. Conditional on h the Y_j are independent (Step 1 of Lemma B.2). Moreover by B3 and the sub-Gaussianity of $z_j^\top h$ we have uniformly bounded moments $\sup_j \mathbb{E}[|Y_j|^p] \leq C_{Y,p} < \infty$ for any fixed p (constants depend on B_0, L and sub-Gaussian norms). Standard moment inequalities for weighted sums of independent sub-Gaussian-type variables (Rosenthal-type bounds) yield, for $p = 4$, a universal constant C_{Ros} such that $\mathbb{E}\left[\left(\sum_{j=1}^m c_j Y_j\right)^4\right] \leq C_{\text{Ros}} \left(\left(\sum_{j=1}^m c_j^2\right)^2 + \sum_{j=1}^m c_j^4\right) \cdot \sup_j \mathbb{E}[Y_j^4]$. Using the scalings $|c_j|^2 = u_j^2 o_j^2 \leq (C_u/m) \cdot (C_o^2/m) = C_u C_o^2/m^2$ we obtain $\sum_{j=1}^m c_j^2 \leq m \cdot (C_u C_o^2/m^2) = C_u C_o^2/m$ and $\sum_{j=1}^m c_j^4 \leq m \cdot (C_u C_o^2/m^2)^2 = O(1/m^3)$. Hence the dominant term is $(\sum_j c_j^2)^2 = O(1/m^2)$. Therefore there exists C_{ud} such that $\mathbb{E}[(u^\top d)^4] \leq C_{ud}/m^2$.

Combining with $\mathbb{E}[r^4] \leq C_r$ we get $\sqrt{\mathbb{E}[r^2 (u^\top d)^2]} \leq C_2/\sqrt{m}$ for some C_2 depending only on the model constants. Thus $|\varepsilon^{(2)}| \leq (C_2 C_h)/\sqrt{m}$, and dividing by $A \geq \tau_n m_0$ yields $\frac{|\varepsilon^{(2)}|}{A} \leq \frac{C_2 C_h}{m_0} \cdot \frac{1}{\tau_n \sqrt{m}}$.

Finally, adding the two contributions and setting $C_{\text{tot}} := \sqrt{\tilde{C}_{rT} C_{\text{var}}} + C_2 C_h$ gives $\frac{|\varepsilon_n|}{A} \leq \frac{C_{\text{tot}}}{m_0}$.
 $\frac{1}{\tau_n \sqrt{m}}$, as claimed. $\tau_n \sim n^{-\frac{1}{6}}$, to be added later, which means $\frac{|\varepsilon_n|}{A} \rightarrow 0$ $\square$

B.5 THEOREM (SINGLE-SPIKE SINGLE-STEP IMPROVEMENT).

Theorem B.2. Assume B1–B5 and suppose $L_0(W)$ is L -smooth in W on a neighborhood of the current parameter. Fix $\vartheta \in (0, 1)$. Suppose Lemma B.3 applies and hence for sufficiently large m we have $|\varepsilon_n| \leq \vartheta A$ (equivalently $\alpha \geq (1 - \vartheta)A$). Choose step-size $\eta = \kappa A$ for sufficiently small $\kappa > 0$ so that higher-order terms are controlled. Assume the spike is active (so the subgradient of L_{spike} equals $g_{\text{spike}}(W) = -2s uv^\top$). Then there exists $c_0 = \frac{3}{4}\kappa s(1 - \vartheta) > 0$ such that for all sufficiently large m , $\Delta_t \leq -c_0 \gamma A^2$, i.e. the spike-augmented update yields an immediate extra population-level decrease in L_0 of order $\Theta(\gamma A^2)$ per step.

Proof. By L -smoothness of L_0 in W we have for any update Δ , $L_0(W + \Delta) \leq L_0(W) + \langle \nabla_W L_0(W), \Delta \rangle + \frac{L}{2} \|\Delta\|_F^2$. Apply this with $\Delta = \Delta_{\text{sp}}$ and $\Delta = \Delta_{\text{van}}$ and subtract the two inequalities to obtain $\Delta_t \leq \langle \nabla_W L_0, \Delta_{\text{sp}} - \Delta_{\text{van}} \rangle + \frac{L}{2} (\|\Delta_{\text{sp}}\|_F^2 - \|\Delta_{\text{van}}\|_F^2)$. Recall $\Delta_{\text{van}} = -\eta \nabla_W L_0$ and $\Delta_{\text{sp}} = -\eta(\nabla_W L_0 + \gamma g_{\text{spike}})$, hence $\Delta_{\text{sp}} - \Delta_{\text{van}} = -\eta \gamma g_{\text{spike}} = 2\eta \gamma s uv^\top$ (since $g_{\text{spike}} = -2s uv^\top$). Therefore the linear term equals $\langle \nabla_W L_0, \Delta_{\text{sp}} - \Delta_{\text{van}} \rangle = 2\eta \gamma s \langle \nabla_W L_0, uv^\top \rangle = -2\eta \gamma s \alpha$. Using Lemma B.3 and the assumption $\alpha \geq (1 - \vartheta)A$ we bound this by $-2\eta \gamma s(1 - \vartheta)A$.

For the quadratic difference we use $\|A + B\|_F^2 - \|A\|_F^2 \leq 2\langle A, B \rangle + \|B\|_F^2$ with $A = -\eta \nabla_W L_0$ and $B = -\eta \gamma g_{\text{spike}}$ to obtain $\|\Delta_{\text{sp}}\|_F^2 - \|\Delta_{\text{van}}\|_F^2 \leq 2\eta^2 \gamma \langle \nabla_W L_0, g_{\text{spike}} \rangle + \eta^2 \gamma^2 \|g_{\text{spike}}\|_F^2$. Using $\|g_{\text{spike}}\|_F = 2s$ and the trivial bound $|\langle \nabla_W L_0, g_{\text{spike}} \rangle| \leq \|\nabla_W L_0\|_F \|g_{\text{spike}}\|_F$ yields $\Delta_t \leq -2\eta \gamma s(1 - \vartheta)A + L\eta^2 \gamma \|\nabla_W L_0\|_F (2s) + \frac{L}{2} \eta^2 \gamma^2 (2s)^2$.

Choose $\eta = \kappa A$ with $\kappa > 0$ small so that the positive η^2 terms are at most half of the leading linear term in magnitude (this is possible since the linear term scales as η and the positive remainder scales as η^2). Concretely pick κ small depending on $L, \gamma, \|\nabla_W L_0\|_F, s, A$ so that $L\eta^2 \gamma \|\nabla_W L_0\|_F (2s) + \frac{L}{2} \eta^2 \gamma^2 (2s)^2 \leq \frac{1}{2} \cdot 2\eta \gamma s(1 - \vartheta)A$. Then $\Delta_t \leq -2\eta \gamma s(1 - \vartheta)A + \frac{1}{2} \cdot 2\eta \gamma s(1 - \vartheta)A = -\frac{3}{2} \eta \gamma s(1 - \vartheta)A$. Substitute $\eta = \kappa A$ to get $\Delta_t \leq -\frac{3}{2} \kappa \gamma s(1 - \vartheta)A^2$. Setting $c_0 := \frac{3}{4} \kappa s(1 - \vartheta)$ proves the claim. $\square$